THYROID DISEASE DETECTION

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PROBLEM STATEMENT

Develop a web-based platform for thyroid disease detection, making it accessible to users for self-assessment and early diagnosis. The platform will provide a user-friendly interface that allows individuals to input relevant health data, and it will use a machine learning model to assess the likelihood of thyroid disease. Users will receive instant feedback on their health status, and if necessary, they will be directed to seek professional medical guidance.

INTRODUCTION

Thyroid disorders are among the most common **endocrine disorders**, affecting millions of people worldwide. Proper diagnosis often requires a combination of **patient history, clinical examination, and specialized blood tests**. However, access to healthcare and timely diagnosis can be challenging for many.

This project aims to bridge that gap by providing a **web-based platform** that uses a machine learning algorithm to **predict the type of thyroid** condition a person might have, based on a questionnaire and specific blood test values. By leveraging technology and data science, this solution offers a preliminary assessment tool to help users gain insights into their thyroid health.

EXISTING SYSTEM AND DISADVANTAGES

1. Accuracy and Reliability Concerns

2. Data Privacy and Security

3. Ethical and Accountability Risks

4. Compliance with Regulations

5. Continuous Maintenance and Improvement

OVERCOMING EXISTING REAL WORLD PROBLEMS

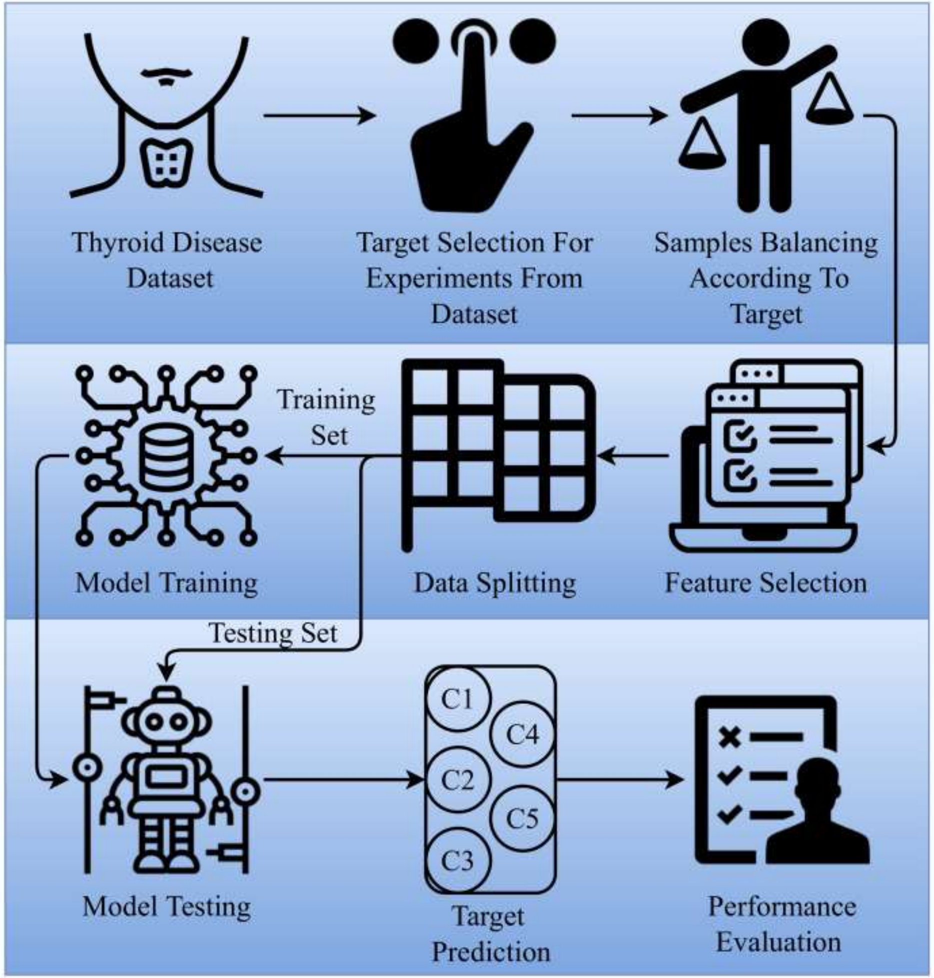
1. ML models can analyze large amount of data leading to more **accuracy**
2. Eliminating the need for **frequent visits** to healthcare centers
3. **Personalized recommendations** based on individual patient data using ML models
4. Keeping track of the user’s **history**
5. They can serve as clinical decision support for health care providers.

PROPOSED SYSTEM WITH ADVANTAGES

- Taking user input 21 questions including blood test values.
- Random forest - ensemble model known for delivering high accuracy in both classification and regression tasks.
- random forests helps mitigate errors by combining multiple models, leading to more consistent and reliable results.
- use a subset of features for each split, reducing the number of computations during the prediction process.
- Random forests can perform predictions in parallel by utilizing multi-core processors

TECHNOLOGY STACK

- ❑ Python
- ❑ Html
- ❑ JavaScript
- ❑ Css (Cascading style sheets)
- ❑ MongoDB
- ❑ Express
- ❑ Node JS



The **targets** for prediction in ML algorithm were:

- **Healthy without thyroid**
- **hyperthyroid**
- **Compensated hypothyroid**
- **Primary hypothyroid**
- **Secondary hypothyroid**

CODING / IMPLEMENTATION

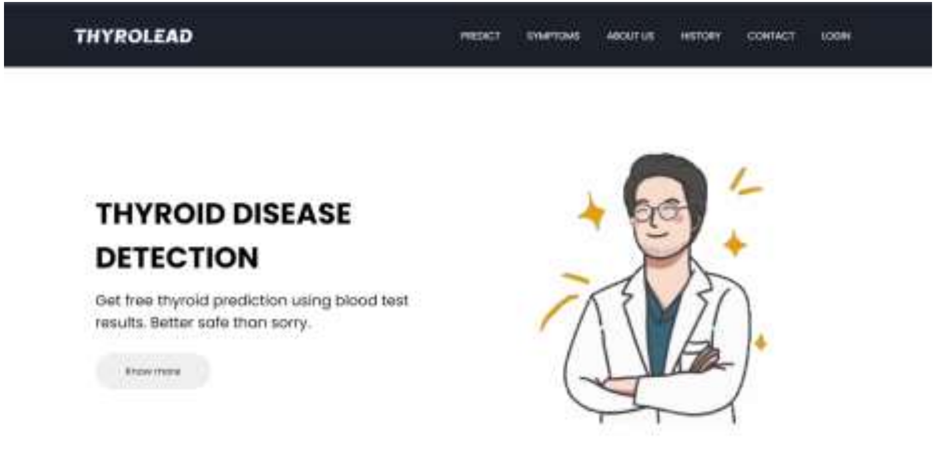
- **classification model** - **100 decision trees** in random forest
- **Express framework** to define app routes and run the website on local server.
- The route for handling **predictions** - **API running on local server 5000**.
- Embedded javascript view engine (ejs)
- Mongodb.js to define with database
- User authentication using session variables
- Server is set to **listen on port 3000**.

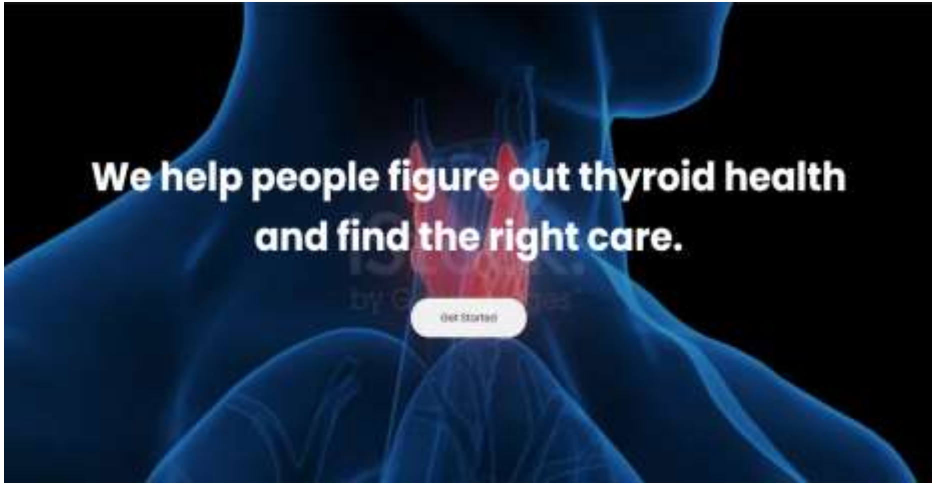
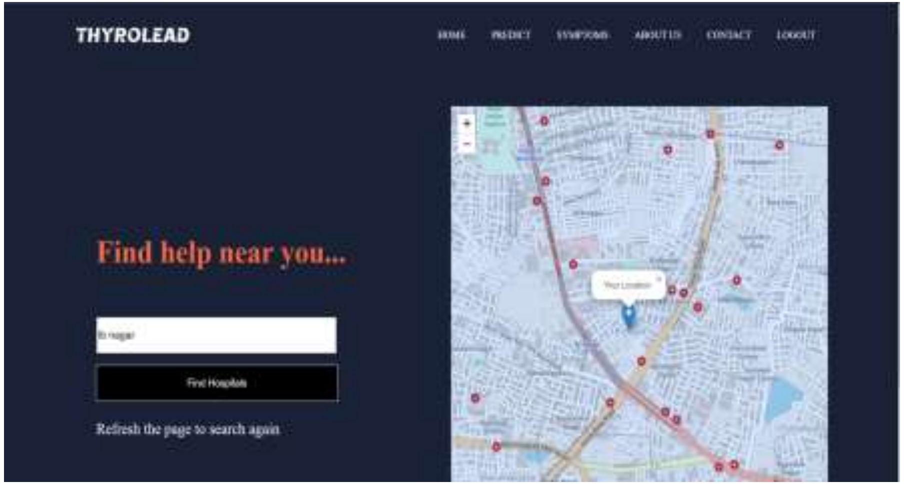
TESTING AND FINAL IMPLEMENTATION

- Dataset contains **10721 rows** of data and **21 columns**.
- Model training **80%** , testing **20%.**
- **Random Forest classifier**
- **98.4%** accuracy and 97.8% precision.
- Nodemon app.js
- MongoDB local database collection - ThyroidDB

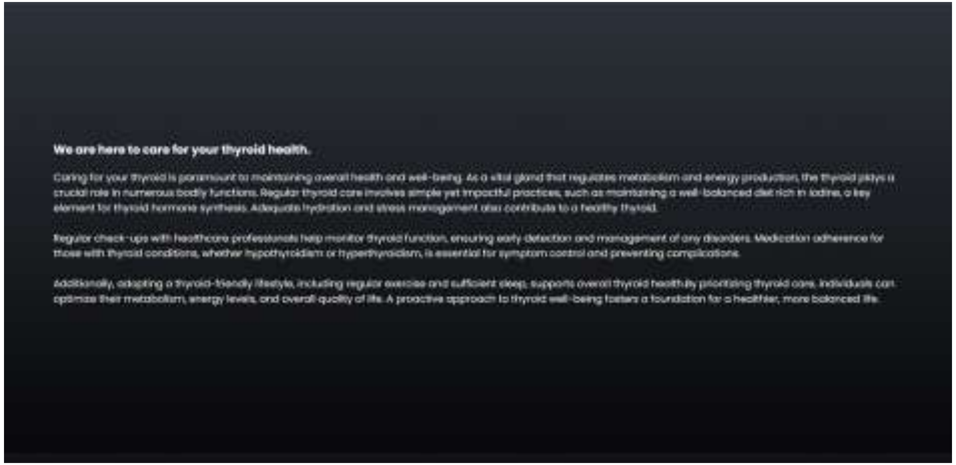
```
Accuracy: 0.9804
Precision: 0.9789
Recall: 0.9804
F1 Score: 0.9796

Confusion Matrix:
[[  38    0    0    0]
 [   0   13   23    0]
 [   1   17 2022    1]
 [   0    0    0   30]]
```


THYROLEAD
PREDICT
SYMPTOMS
ABOUT US
HISTORY
CONTACT
LOGIN
THYROID DISEASE DETECTION
Get free thyroid prediction using blood test results. Better safe than sorry.
Know more

Here's how we help..
STEP 1
STEP 2
STEP 3

We help people figure out thyroid health and find the right care.
Get Started

THYROLEAD
HOME
PREDICT
SYMPTOMS
ABOUT US
CONTACT
LOGOUT
Find help near you...
Find Hospitals
Refresh the page to search again


Here's how we help:
1. Use our A.I to predict your thyroid health.
1. Find nearby healthcare professionals for further diagnosis and treatment.
?
?


We are here to care for your thyroid health.
Caring for your thyroid is paramount to maintaining overall health and well-being. As a vital gland that regulates metabolism and energy production, the thyroid plays a crucial role in numerous bodily functions. Regular thyroid care involves simple yet impactful practices, such as maintaining a well-balanced diet rich in iodine, a key element for thyroid hormone synthesis. Adequate hydration and stress management also contribute to a healthy thyroid.
Regular check-ups with healthcare professionals help monitor thyroid function, ensuring early detection and management of any disorders. Medication adherence for those with thyroid conditions, whether hypothyroidism or hyperthyroidism, is essential for symptom control and preventing complications.
Additionally, adopting a thyroid-friendly lifestyle, including regular exercise and sufficient sleep, supports overall thyroid health.By prioritizing thyroid care, individuals can optimize their metabolism, energy levels, and overall quality of life. A proactive approach to thyroid well-being fosters a foundation for a healthier, more balanced life.


Registration
username
name
password
I agree to the terms & conditions
Register
Already have an account? Login


Login
username
password
login
Don't have an account? Register

What is Thyroid?

The thyroid gland is an important organ of the endocrine system. It is located at the front of the neck just above where your collarbones meet. The gland makes the hormones that control the way every cell in the body uses energy. This process is called metabolism.

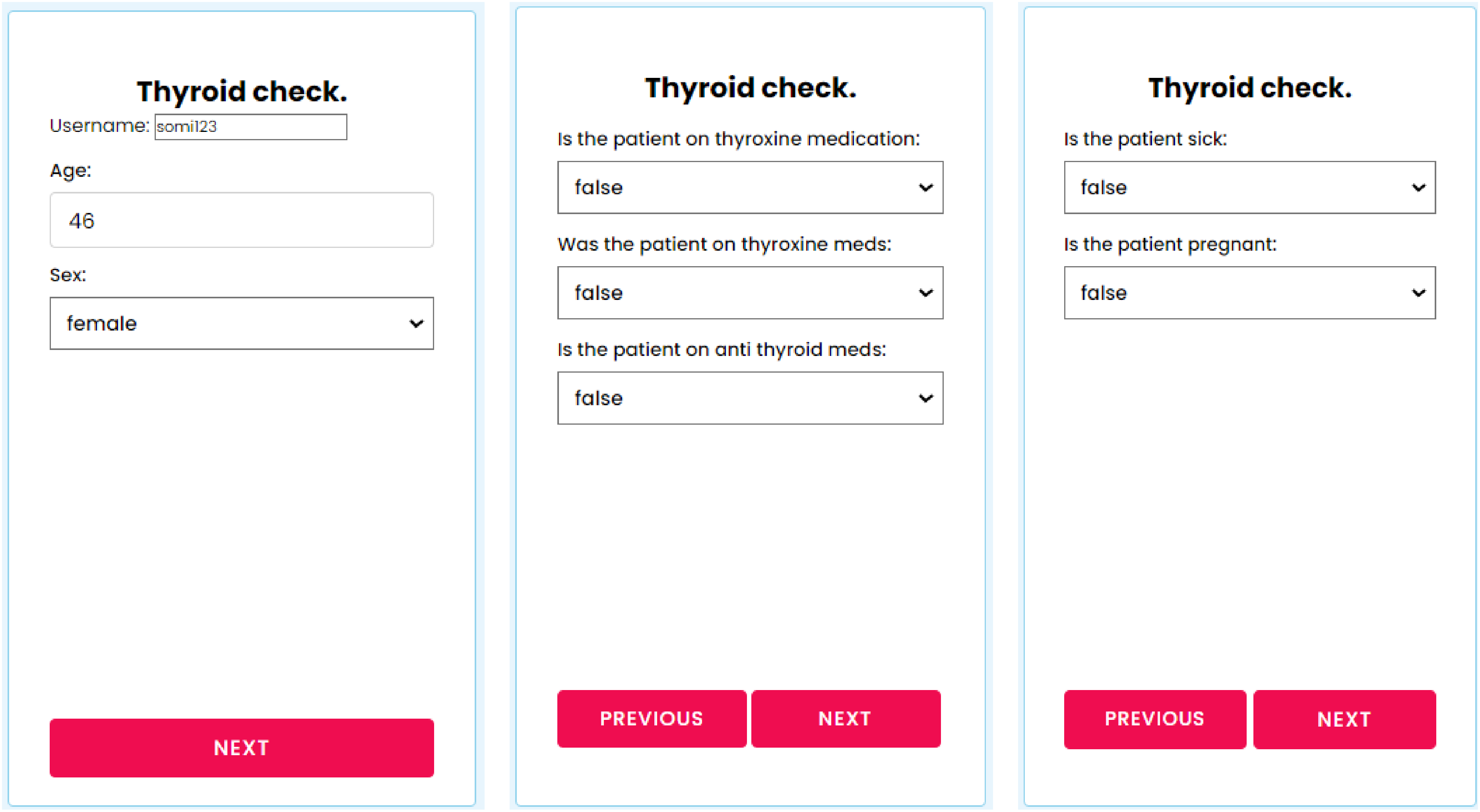
Symptoms of an underactive thyroid (hypothyroidism) can include:

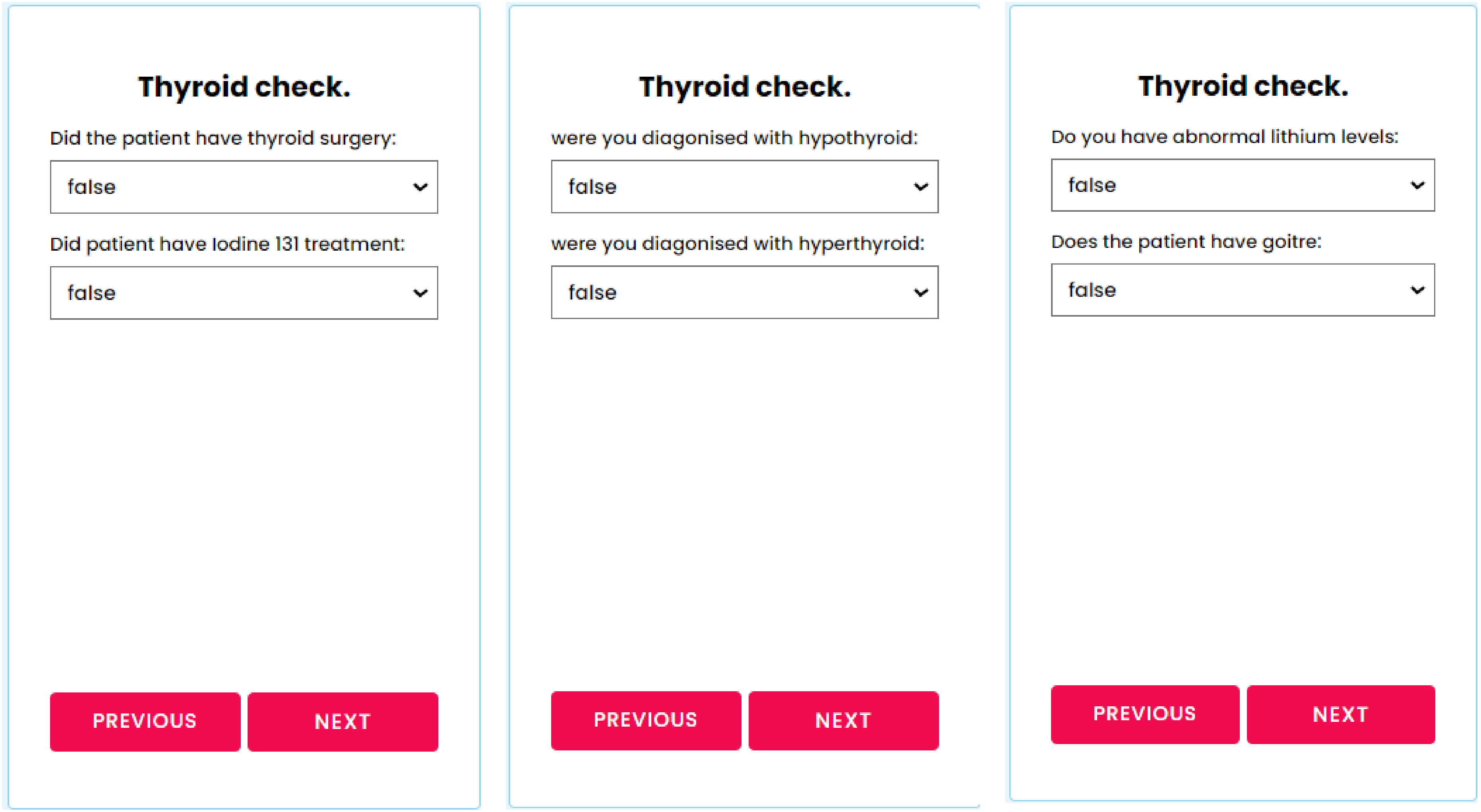
What are the common symptoms of thyroid disease?

There are a variety of symptoms you could experience if you have a thyroid disease. Unfortunately, symptoms of a thyroid condition are often very similar to the signs of other medical conditions and stages of life. This can make it difficult to know if your symptoms are related to a thyroid issue or something else entirely. For the most part, the symptoms of thyroid disease can be divided into two groups — those related to having too much thyroid hormone (hyperthyroidism) and those related to having too little thyroid hormone (hypothyroidism).

The specific blood tests that will be done to test your thyroid can include:

- **Thyroid-stimulating hormone (TSH)** is produced in the pituitary gland and regulates the balance of thyroid hormones — including T4 and T3 — in the bloodstream. This is usually the first test your provider will do to check for thyroid hormone imbalance. Most of the time, thyroid hormone deficiency (hypothyroidism) is associated with an elevated TSH level, while thyroid hormone excess (hyperthyroidism) is associated with a low TSH level. If TSH is abnormal, measurement of thyroid hormones directly, including thyroxine (T4) and triiodothyronine (T3) may be done to further evaluate the problem. Normal TSH range for an adult: 0.40 - 4.50 mIU/mL (milli-international units per liter of blood).
- **T4:** Thyroxine tests for hypothyroidism and hyperthyroidism, and used to monitor treatment of thyroid disorders. Low T4 is seen with hypothyroidism, whereas high T4 levels may indicate hyperthyroidism. Normal T4 range for an adult: 5.0 – 11.0 ug/dL (micrograms per deciliter of blood).
- **FT4:** Free T4 or free thyroxine is a method of measuring T4 that eliminates the effect of proteins that naturally bind T4 and may prevent accurate measurement. Normal FT4 range for an adult: 0.9 - 1.7 ng/dL (nanograms per deciliter of blood)
- **T3:** Triiodothyronine tests help diagnose hyperthyroidism or to show the severity of hyperthyroidism. Low T3 levels can be observed in hypothyroidism, but more often this test is useful in the diagnosis and


Thyroid check.
Username: somi123
Age:
46
Sex:
female
NEXT
Thyroid check.
Is the patient on thyroxine medication:
false
Was the patient on thyroxine meds:
false
Is the patient on anti thyroid meds:
false
PREVIOUS
NEXT
Thyroid check.
Is the patient sick:
false
Is the patient pregnant:
false
PREVIOUS
NEXT


Thyroid check.
Did the patient have thyroid surgery:
false
Did patient have Iodine 131 treatment:
false
PREVIOUS
NEXT
Thyroid check.
were you diagonised with hypothyroid:
false
were you diagonised with hyperthyroid:
false
PREVIOUS
NEXT
Thyroid check.
Do you have abnormal lithium levels:
false
Does the patient have goitre:
false
PREVIOUS
NEXT

Thyroid check.

Does the patient have tumor:

false

Do you have hypopituitarism:

false

Is the patient psychologically unstable:

false

PREVIOUS NEXT

Thyroid check.

TSH (Thyroid stimulating hormone):

0.4 - 4.0 mU/L

T3 (Triiodothyronine):

80 - 200 ng/DL

TT4 (Total Thyroxine):

4.5 - 12.5 μU/dL

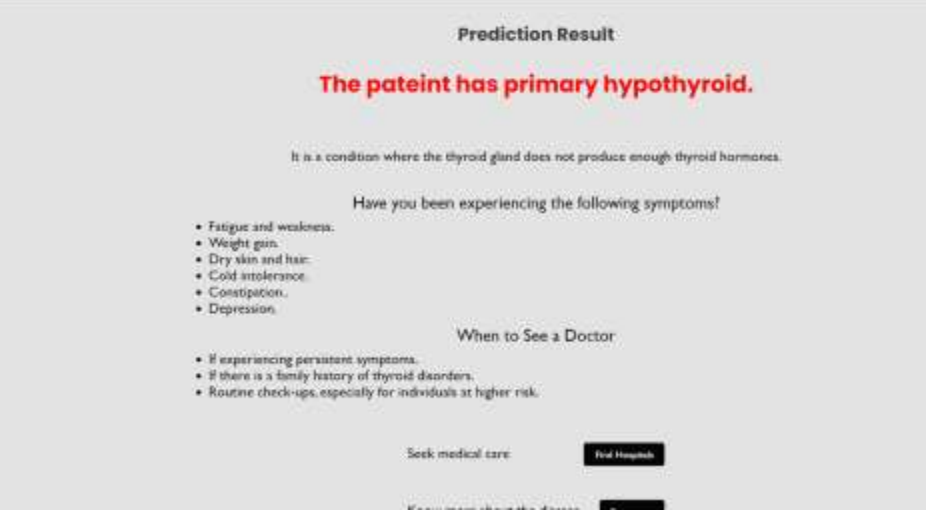
T4U (Thyroxine uptake):

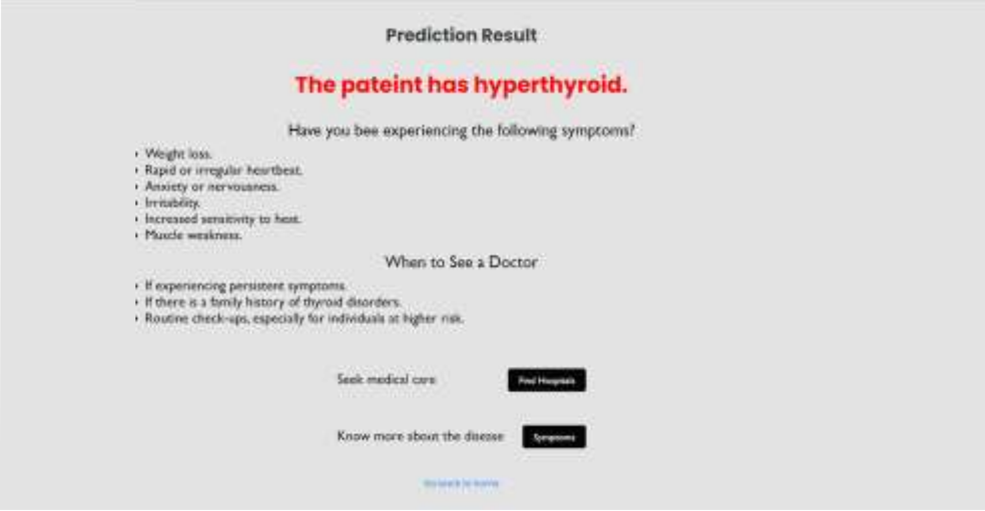
0.9 - 1.1

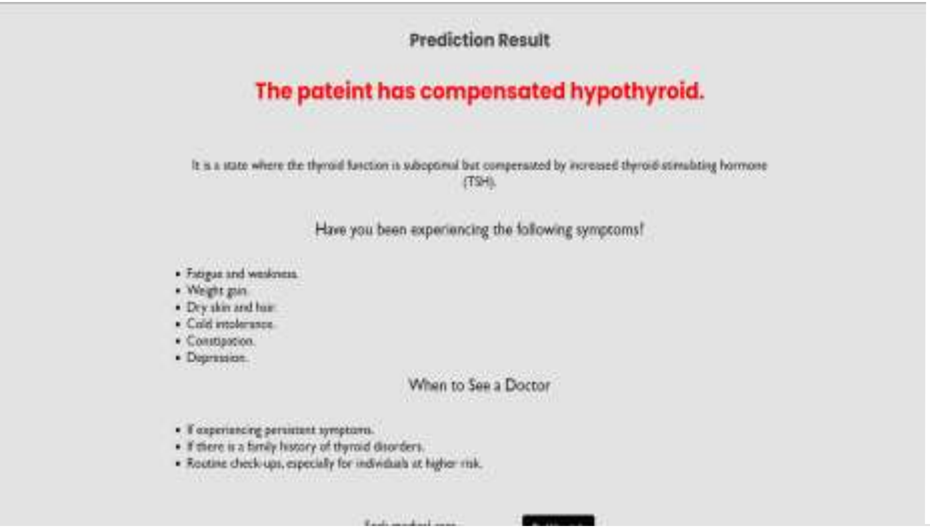
FTI (Free thyroxine index):

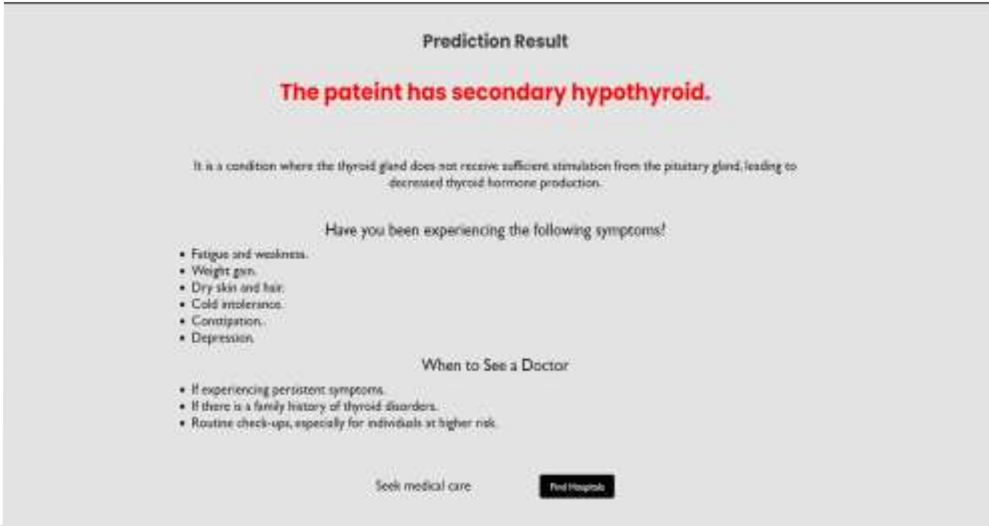
5 - 11

PREVIOUS SUBMIT


Prediction Result
The pateint has primary hypothyroid.
It is a condition where the thyroid gland does not produce enough thyroid hormones.
Have you been experiencing the following symptoms?
Fatigue and weakness.
Weight gain.
Dry skin and hair.
Cold intolerance.
Constipation.
Depression.
When to See a Doctor
If experiencing persistent symptoms.
If there is a family history of thyroid disorders.
Routine check-ups, especially for individuals at higher risk.
Seek medical care
Find Hospitals


Prediction Result
The pateint has hyperthyroid.
Have you bee experiencing the following symptoms?
Weight loss.
Rapid or irregular heartbeat.
Anxiety or nervousness.
Irritability.
Increased sensitivity to heat.
Muscle weakness.
When to See a Doctor
If experiencing persistent symptoms.
If there is a family history of thyroid disorders.
Routine check-ups, especially for individuals at higher risk.
Seek medical care
Find Hospitals
Know more about the disease
Symptoms


Prediction Result
The pateint has compensated hypothyroid.
It is a state where the thyroid function is suboptimal but compensated by increased thyroid-stimulating hormone (TSH).
Have you been experiencing the following symptoms?
Fatigue and weakness.
Weight gain.
Dry skin and hair.
Cold intolerance.
Constipation.
Depression.
When to See a Doctor
If experiencing persistent symptoms.
If there is a family history of thyroid disorders.
Routine check-ups, especially for individuals at higher risk.


Prediction Result
The pateint has secondary hypothyroid.
It is a condition where the thyroid gland does not receive sufficient stimulation from the pituitary gland, leading to decreased thyroid hormone production.
Have you been experiencing the following symptoms?
Fatigue and weakness.
Weight gain.
Dry skin and hair.
Cold intolerance.
Constipation.
Depression.
When to See a Doctor
If experiencing persistent symptoms.
If there is a family history of thyroid disorders.
Routine check-ups, especially for individuals at higher risk.
Seek medical care
Find Hospitals

History Page

User: somi123

Message:

Patient is healthy without thyroid.

Timestamp: Wed Jan 24 2024 19:58:41 GMT+0530 (India Standard Time)

Message:

Patient has hyperthyroid.

Timestamp: Wed Jan 24 2024 20:04:53 GMT+0530 (India Standard Time)

Message:

Patient is healthy without thyroid.

Timestamp: Wed Jan 24 2024 20:41:08 GMT+0530 (India Standard Time)

Message:

Patient is healthy without thyroid.

Timestamp: Wed Jan 24 2024 20:43:06 GMT+0530 (India Standard Time)

Message:

Patient is healthy without thyroid.

Timestamp: Wed Jan 24 2024 20:43:25 GMT+0530 (India Standard Time)

Message:

Patient is healthy without thyroid

CONCLUSION AND FUTURE SCOPE

1. Provides preliminary indication of health status.
2. Help bridge the gap for accessible healthcare resources.
3. Ensuring robust data privacy, and resolving ethical concerns.
4. Helps reduce the workload for healthcare professionals, allowing them to focus on cases requiring more detailed attention.
5. Can explore integrating with medical systems, expanding diagnostic capabilities, and creating a mobile app to increase accessibility and user engagement.

THANK YOU